

Dataset

Data Source

- Data Source: Google Big Query E-commerce store
- Time: 2019-2026 Records: 181,184 Transactions

Preview of Bigquery Raw Data

Data Cleaning & Transformation

-SQL QUERIES IN BIGQUERY -STEP 1: Understand the Data Model -Main joins:

- `order_items.product_id = products.id`
- `order_items.order_id = orders.order_id`
- `products.distribution_center_id = distribution_centers.id` **Core main table (fact):**
order_items (this drives revenue) Dimension tables:
- products
- distribution_centers
- orders

STEP 2: Create a Base Table

This is how I structure a clean analysis layer: pulling and joining the table WITH base AS (
SELECT oi.order_id, oi.product_id, oi.sale_price, oi.status, oi.created_at,

p.cost,

p.category,

p.department,

p.distribution_center_id,

dc.name AS distribution_center_name

FROM bigquery-public-data.thelook_ecommerce.order_items oi

LEFT JOIN bigquery-public-data.thelook_ecommerce.products p ON oi.product_id = p.id

LEFT JOIN bigquery-public-data.thelook_ecommerce.distribution_centers dc ON
p.distribution_center_id = dc.id)

SELECT * FROM base LIMIT 100;

Data CLEANING

Clean → Transform → Validate

Goal: Remove incorrect, duplicated, or unusable data.

- **Check for NULL Values**

Start by checking critical fields -SELECT COUNT(*) AS total_rows, COUNTIF(order_id IS NULL) AS null_orders, COUNTIF(product_id IS NULL) AS null_products, COUNTIF(sale_price IS NULL) AS null_sale_price, COUNTIF(cost IS NULL) AS null_cost FROM base;

Remove Invalid Prices

Check for negative or zero revenue:

- SELECT * FROM base WHERE sale_price <= 0;

Check for Duplicates

Since order_items is the fact table, duplicates could exist.

- SELECT order_id, product_id, COUNT() AS cnt FROM base GROUP BY order_id, product_id HAVING COUNT() > 1;

Check Status Values

- SELECT DISTINCT status FROM base; **Standardize values like:**
- Returned
- Shipped
- Delivered
- Cancelled

Data Transformation

Create Revenue & Profit Columns Instead of recalculating every time:

- SELECT *, sale_price AS revenue, (sale_price - cost) AS profit, SAFE_DIVIDE(sale_price - cost, sale_price) AS margin FROM base;

Data Validation

Validate Revenue Compare original main table vs cleaned table.

- `SELECT SUM(sale_price) FROM bigquery-public-data.thelook_ecommerce.order_items;` Then compare with your cleaned base:
- `SELECT SUM(revenue) FROM final_clean_table;` If huge difference → something is wrong.

Validate Order Counts

- `SELECT COUNT(DISTINCT order_id) FROM final_clean_table;` Compare with:
- `SELECT COUNT(DISTINCT order_id) FROM bigquery-public-data.thelook_ecommerce.orders;` They should align (except cancelled logic).

Validate Profit Logic

Check if any weird numbers:

- `SELECT * FROM final_clean_table WHERE profit > 10000 OR profit < -10000;` Look for anomalies.

Validate Distribution Centers

- `SELECT distribution_center_name, COUNT(*) FROM final_clean_table GROUP BY distribution_center_name;`

CREATE OR REPLACE TABLE your_dataset.ecommerce_cleaned AS and exported directly into power Bi

```
SELECT order_id, product_id, distribution_center_name, category, department, State, City,
Product, Category, sale_price AS revenue, cost, (sale_price - cost) AS profit,
SAFE_DIVIDE(sale_price - cost, sale_price) AS margin, DATE(created_at) AS order_date,
EXTRACT(YEAR FROM created_at) AS year, FORMAT_DATE('%Y-%m', DATE(created_at)) AS
year_month, status, CASE WHEN status = 'Returned' THEN 1 ELSE 0 END AS is_returned
FROM base WHERE sale_price > 0 AND order_id IS NOT NULL;
```

Create temporary table in bigquery and rename it (MAIN). Export directly to power Bi